

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZeroes(int arr[], int low, int high, int n) {
4      if (low > high)
5          return 0;
6
7      int mid = low + (high - low) / 2;
8
9      if (arr[mid] == 0 && (mid == 0 || arr[mid - 1] == 1))
10         return n - mid;
11
12     if (arr[mid] == 1)
13         return countZeroes(arr, mid + 1, high, n);
14
15     return countZeroes(arr, low, mid - 1, n);
16 }
17
18 int main() {
19     int m;
20     scanf("%d", &m);
21
22     int arr[m];
23
24     for (int i = 0; i < m; i++)
25         scanf("%d", &arr[i]);
26
27     int result = countZeroes(arr, 0, m - 1, m);
28
29     printf("%d\n", result);
30
31     return 0;
32 }

```

	Input	Expected	Got	
✓	5	2	2	✓
	1			
	1			
	1			
	0			
	0			

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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